

HACK THE BOX SHERLOCK - DFIR

AOTR 4: A DRONE IN THE SNOW

Sherlock Scenario

Advent of The Relics 4 - A Drone in the Snow

Read the campaign introduction and supporting information at <https://github.com/hackthebox/advent-of-the-relics>

<https://github.com/hackthebox/advent-of-the-relics>

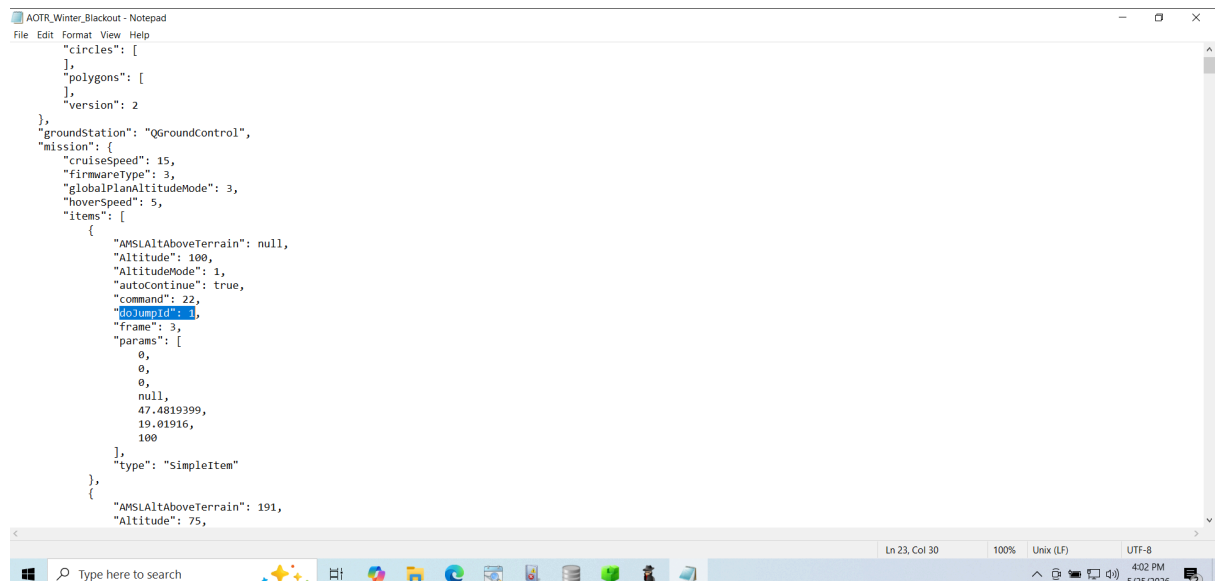
A recovered disk blew the case open. The files inside weren't theory, they were a checklist. Hours later, an unmarked matte-black drone was found half-buried in snow in Budapest, matching the recovered plans too closely to ignore. The drone is now in your lab and you've been given its mission plan and flight telemetry. Reconstruct the route, pinpoint key behavior, and prove what happened before impact, you're running out of time before the real heist begins.

The scenario portrayed in this challenge is entirely fictional and created solely for educational and entertainment purposes. Any resemblance to actual persons, living or dead, organizations, or real events is purely coincidental and unintentional. All characters, scenarios, and data presented are products of imagination.

	Name	Date modified	Type	Size
ss	AOTR_Winter_Blackout.plan	1/6/2026 3:12 AM	PLAN File	26 KB
ds	log.bin	1/6/2026 3:42 AM	BIN File	37,454 KB

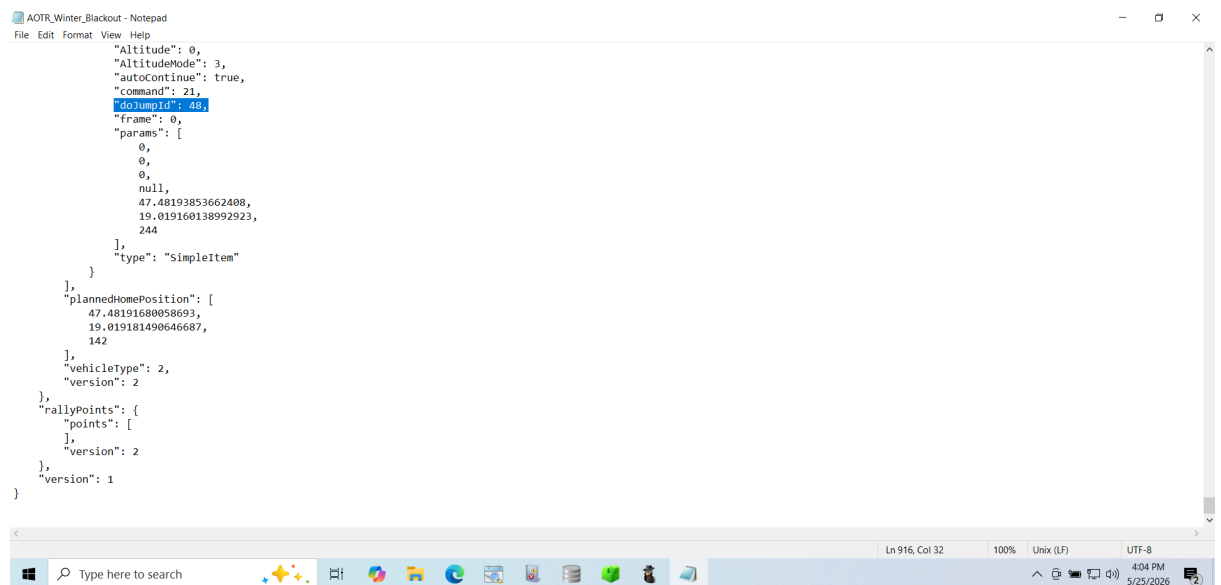
After unzipping the zip file from the Sherlock there are 2 files provided which is a .plan and .bin file.

1. How many total mission items are defined in the flight plan, including Home, Takeoff, and Land commands?



```
File Edit Format View Help
AOTR_Winter_Blackout - Notepad
{
  "circles": [
  ],
  "polygons": [
  ],
  "version": 2
},
"groundStation": "QGroundControl",
"mission": {
  "cruiseSpeed": 15,
  "firmwareType": 3,
  "globalPlanAltitudeMode": 3,
  "hoverSpeed": 5,
  "items": [
    {
      "AMSLAltAboveTerrain": null,
      "Altitude": 100,
      "AltitudeMode": 1,
      "autoContinue": true,
      "command": 22,
      "doJumpId": 1,
      "frame": 3,
      "params": [
        0,
        0,
        0,
        null,
        47.4819399,
        19.01916,
        100
      ],
      "type": "SimpleItem"
    },
    {
      "AMSLAltAboveTerrain": 191,
      "Altitude": 75,
```

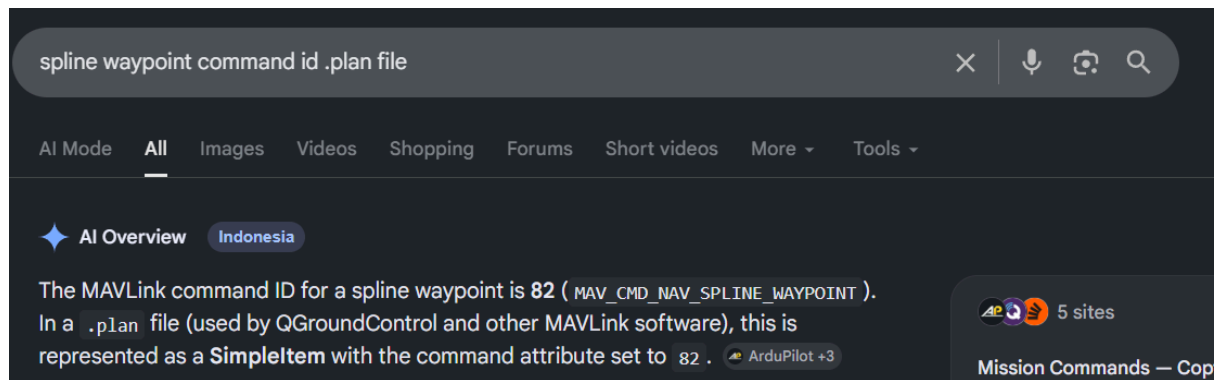
Here opening the AOTR_Winter_Blackout.plan file in Notepad we see from the item it begins from a Jump ID of 1.



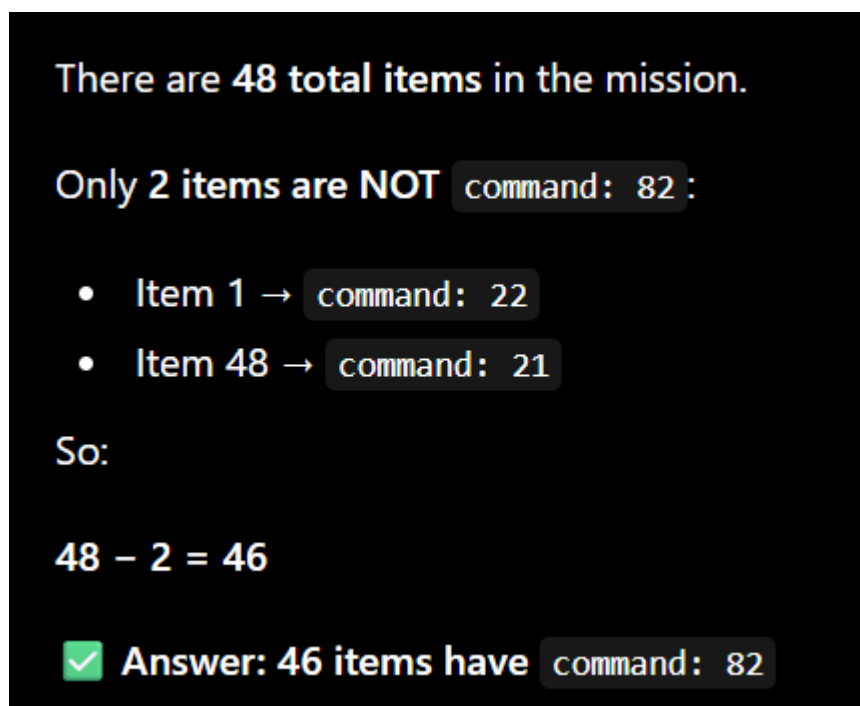
```
      "Altitude": 0,
      "AltitudeMode": 3,
      "autoContinue": true,
      "command": 21,
      "doJumpId": 48,
      "frame": 0,
      "params": [
        0,
        0,
        0,
        null,
        47.48193853662408,
        19.019160138992923,
        244
      ],
      "type": "SimpleItem"
    },
    }
  ],
  "plannedHomePosition": [
    47.48191680058693,
    19.019181490646687,
    142
  ],
  "vehicleType": 2,
  "version": 2
},
"rallyPoints": {
  "points": [
  ],
  "version": 2
},
"version": 1
}
```

And it ends at Jump ID 48, so there is already 48 items from there. Adding them with the plannedHomePosition adds 1 more, so then the answer is **49**

2. How many spline waypoints are in the mission?



Spline Waypoint command ID is 82. Therefore, I will copy paste the whole .plan file and ask ChatGPT to count it.



Therefore among the 48 total items only 2 are not not 82, therefore subtracting 48 by 2 gets **46**

3. Which landmark/building does the drone fly around during the planned route?



Here opening the .plan file on QGroundControl we see the drone fly around a building called the **Citadella**

4. At which mission item did the drone have a planned hold/loiter?



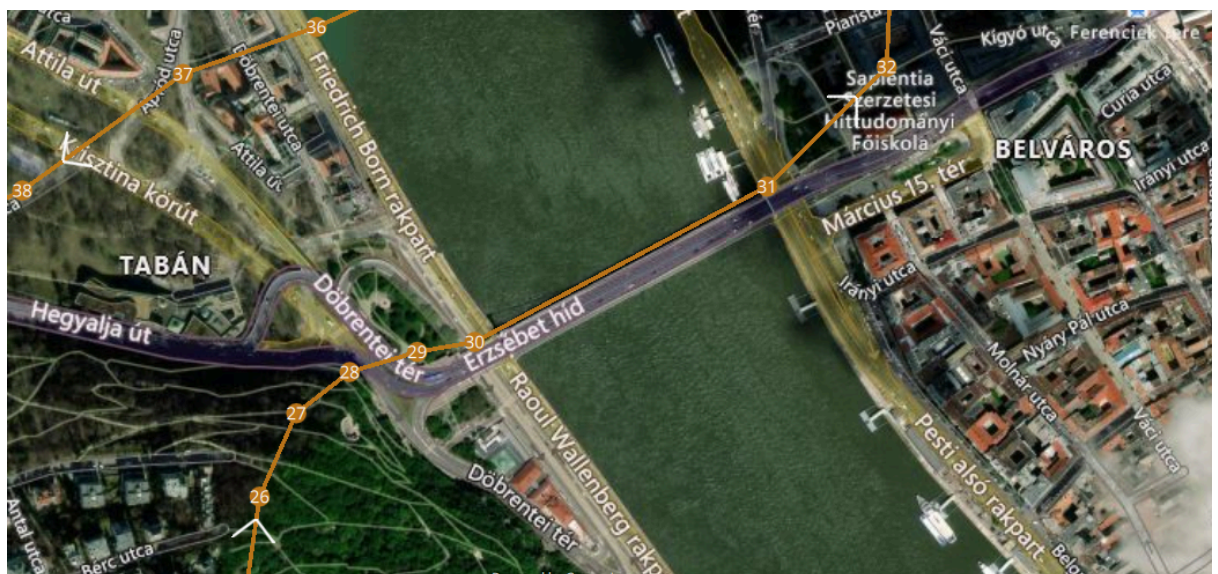
Here we see in the mission item number **18** the drone had a planned hold which lasted for 30 seconds

5. How long was the planned hold time at that mission item (seconds)?

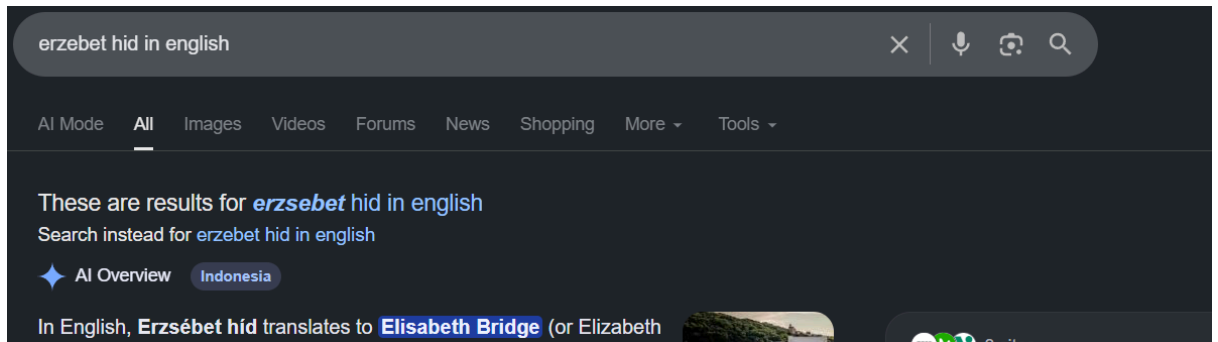


Also can be seen from the previous question's screenshot and was previously mentioned in the previous question that the hold lasted for **30** seconds.

6. Which bridge is the drone planned to pass by while crossing the Danube in order to reach the other part of the city? (English name)



Here I see that there is a bridge called "Erzsébet híd" but then the question asked it to be in English, therefore translating it we get **Elisabeth Bridge**



7. What is the total flight time from takeoff to crash?

```
C:\Users\Rafael.WINDOWS10\Desktop\HacktheBox Sherlock\AOTR4>python mavlogdump.py log.bin --show-types
BAT
IMU
RFND
RCIN
DSF
```

For this question I have to install [mavlogdump.py](#) from [pymavlink/tools/mavlogdump.py at master · ArduPilot/pymavlink · GitHub](#) to analyze the [log.bin](#) file that contains the events. Then I used the `--show-types` to list the tables

```
ESC
MSG
XKQ
FMT
MOTR
```

And then there is MSG, which is human readable text. So I went deeper into that specific table.

```
C:\Users\Rafael.WINDOWS10\Desktop\HacktheBox Sherlock\AOTR4>python mavlogdump.py --type MSG log.bin
2026-01-06 03:16:07.13: MSG {TimeUS : 23319002, Message : ArduCopter V4.4.4 (865cffa5)}
2026-01-06 03:16:07.13: MSG {TimeUS : 23319002, Message : 9f407028b696}
2026-01-06 03:16:07.13: MSG {TimeUS : 23319002, Message : Param space used: 210/4096}
2026-01-06 03:16:07.13: MSG {TimeUS : 23319002, Message : RC Protocol: SITL}
2026-01-06 03:16:07.13: MSG {TimeUS : 23319002, Message : New mission}
2026-01-06 03:16:07.13: MSG {TimeUS : 23319002, Message : New rally}
2026-01-06 03:16:07.13: MSG {TimeUS : 23319002, Message : New fence}
2026-01-06 03:16:07.13: MSG {TimeUS : 23319002, Message : Frame: QUAD/X}
2026-01-06 03:16:07.13: MSG {TimeUS : 23319002, Message : GPS 1: detected as u-blox at 230400 baud}
2026-01-06 03:16:13.25: MSG {TimeUS : 29434055, Message : Mode change to GUIDED failed: requires position}
2026-01-06 03:16:24.17: MSG {TimeUS : 40358850, Message : Mode change to GUIDED failed: requires position}
2026-01-06 03:16:25.22: MSG {TimeUS : 41408430, Message : EKF3 IMU1 is using GPS}
2026-01-06 03:16:25.23: MSG {TimeUS : 41418426, Message : EKF3 IMU0 is using GPS}
2026-01-06 03:16:32.70: MSG {TimeUS : 48893768, Message : Mission: 1 Takeoff}
2026-01-06 03:16:35.22: MSG {TimeUS : 51408595, Message : EKF3 IMU0 MAG0 in-flight yaw alignment complete}
```

Then in there I found the takeoff time.

```
2026-01-06 03:26:59.84: MSG {TimeUS : 676028647, Message : Mission: 1 Takeoff}
2026-01-06 03:27:11.62: MSG {TimeUS : 687813098, Message : SIM Hit ground at 12.93254 m/s}
2026-01-06 03:28:44.40: MSG {TimeUS : 780593471, Message : ArduCopter V4.4.4 (865cffa5)}
2026-01-06 03:28:44.43: MSG {TimeUS : 780614296, Message : 9f407028b696}
2026-01-06 03:28:44.43: MSG {TimeUS : 780614296, Message : Frame: QUAD/X}
```

And here, it is also stated when it hit the ground subtracting the time we get $687813098 - 48893768 = 638919330$.

But that is in milliseconds, converting it to the format asked by the question we get:

Step 2: Convert to seconds

These timestamps are in **microseconds (μs)**.

So divide by 1,000,000:

$$638,919,330 / 1,000,000 = 638.91933 \text{ seconds}$$

Step 3: Convert to hh:mm:ss.sss

- Minutes: $638 \div 60 = 10$ minutes
- Remaining seconds: $638 - 600 = 38.91933$

So:

$$00:10:38.919$$

Therefore the answer is **00:10:38.919**

8. What is the exact log timestamp of the crash event (TimeUS)?

```
2026-01-06 03:26:34.57: MSG {TimeUS : 650761258, Message : Reached command #27}
2026-01-06 03:26:34.57: MSG {TimeUS : 650761258, Message : Mission: 28 SplineWP}
2026-01-06 03:26:49.01: MSG {TimeUS : 665203812, Message : Reached command #28}
2026-01-06 03:26:49.01: MSG {TimeUS : 665203812, Message : Mission: 29 SplineWP}
2026-01-06 03:26:59.84: MSG {TimeUS : 676026148, Message : Disarming motors}
2026-01-06 03:26:59.84: MSG {TimeUS : 676028647, Message : Mission: 1 Takeoff}
2026-01-06 03:27:11.62: MSG {TimeUS : 687813098, Message : SIM Hit ground at 12.93254 m/s}
2026-01-06 03:28:44.40: MSG {TimeUS : 780593471, Message : ArduCopter V4.4.4 (865cffa5)}
2026-01-06 03:28:44.43: MSG {TimeUS : 780614296, Message : 9f407028b696}
2026-01-06 03:28:44.43: MSG {TimeUS : 780614296, Message : Frame: QUAD/X}
```

The exact timestamp when the drone crashed was used to find the solution to the previous question and it is **687813098**.

9. What are the coordinates where the drone crashed (lat, lon)?

```
(venv)-(rafael@raf)-[~/HTB_SHERLOCKS/AOTR4]
$ python3 mavlogdump.py log.bin | grep -B30 687813098 | grep Lat
2026-01-06 04:27:11.59: GPS {TimeUS : 687783943, I : 0, Status : 6, GMS : 160049600, GWk : 2400, NSats : 10, HDop : 1.21, Lat : 47.4903055, Lng : 19.0460476, Alt : 121.76, Spd : 3.207000255584717, GCrs : 294.954833984375, VZ : 12.932000160217285, Yaw : 0.0, U : 1}
```

Some questions onwards will have solution using Kali Linux since grep is faster than Select-String on Powershell or findstr on Command Prompt, so I just got from log.bin the string 687813098 but get the 30 lines before

that and get the line where contains "Lat" and the question asked in the format of Lat, Lng, so the answer is **47.4903055, 19.0460476**

10. What was the ground impact speed reported at the moment of crash (m/s)?

```
2026-01-06 03:26:34.57: MSG {TimeUS : 650761258, Message : Mission: 28 SplineWP}
2026-01-06 03:26:49.01: MSG {TimeUS : 665203812, Message : Reached command #28}
2026-01-06 03:26:49.01: MSG {TimeUS : 665203812, Message : Mission: 29 SplineWP}
2026-01-06 03:26:59.84: MSG {TimeUS : 676026148, Message : Disarming motors}
2026-01-06 03:26:59.84: MSG {TimeUS : 676028647, Message : Mission: 1 Takeoff}
2026-01-06 03:27:11.62: MSG {TimeUS : 687813098, Message : SIM Hit ground at 12.93254 m/s}
2026-01-06 03:28:44.40: MSG {TimeUS : 780593471, Message : ArduCopter V4.4.4 (865cffa5)}
2026-01-06 03:28:44.43: MSG {TimeUS : 780614296, Message : 9f407028b696}
2026-01-06 03:28:44.43: MSG {TimeUS : 780614296, Message : Frame: QUAD/X}
```

Here in the screenshot above, it is stated that the Drone hit the ground at **12.93254** m/s

11. What was the maximum GPS altitude reached during the flight (meters)?

```
(venv)-(rafael@raf)-[~/HTB_SHERLOCKS/A0TR4]
$ python3 mavlogdump.py --type GPS log.bin \
| grep -o "Alt[^\,]*" \
| sort -n \
| tail -1
Alt : 377.07
```

For this question, I grep again using the GPS table and then I grep the strings with the Alt string and a comma, and then sort it and the tail -1 is to get the highest value since it is the maximum **377.07**

12. What was the fastest GPS ground speed recorded (m/s)?

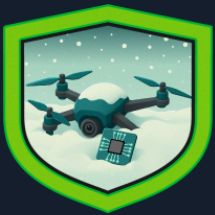
```
(venv)-(rafael@raf)-[~/HTB_SHERLOCKS/A0TR4]
$ python3 mavlogdump.py --type GPS log.bin \
| grep -o "Spd : [0-9.]*" \
| cut -d' ' -f3 \
| sort -n \
| tail -1
10.24000072479248
```


Here I searched again on the GPS table this time I filter with "Spd" with colon and regex for numbers and the cut makes it so it only extracts the number, then I sort it and get the last value since it is asking for the fastest which would be the biggest as well so the answer is **10.24**

13. What are the coordinates where the drone took off from (lat, lon), and therefore a good location for the police raid to catch the gang member?

```
(venv)-(rafael@raf)-[~/HTB_SHERLOCKS/AOTR4]
$ python3 mavlogdump.py --type GPS log.bin \
| sed -nE 's/.*Lat : ([0-9.-]+), Lng : ([0-9.-]+).*/\1, \2/p' \
| head -1
47.4819399, 19.01916
```

Here for the last question, I just filter for the first last question I just filter using the strings Lat and Lng and the regex for numerics again and still on the GPS table at log.bin and just filtering it to only 1 entry by head -1. So the answer is **47.4819399, 19.0191600**



You have solved AoTR 4: A Drone in the Snow!

Congratulations **rafaelangelo** best of luck in capturing flags ahead!

#308	26 May 2026	Active
Sherlock Rank	Solve Date	Sherlock State

Ok Share